PRIME DIVISORS MET BY A MEROMORPHIC SECTION FORM A LOCALLY FINITE SET

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1158-1165, immediately following the statement at 1140-1156.

Editorial note. One same-chapter lemma stub is recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Meromorphic sections are honest sections on a dense open. definition of the meromorphic sections module K_X(L) invoked at proof lines 1158-1160. s corresponds to an actual section of L over some nonempty open U where it generates L.
- Local finiteness of irreducible components of the complement. Lemma lemma-components-locally-finite of this chapter, cited at proof lines 1164-1165. Irreducible components of |X| setminus |U| form a locally finite family – the terminal input.

Source reference. Original lemma 0EE5: sources/stacks-spaces-over-fields/pinned-original-source.tex, source lines 725–725. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a locally Noetherian integral algebraic space over S. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $s \in \mathcal{K}_X(\mathcal{L})$ be a regular (i.e., nonzero) meromorphic section of $\mathcal{L}$. Then the sets*

$$\{Z \subset X \mid Z \text{ a prime divisor with generic point } \xi \text{ and } s \text{ not in } \mathcal{L}_\xi\}$$

and

$$\{Z \subset X \mid Z \text{ is a prime divisor and } ord_{Z,\mathcal{L}}(s) \neq 0\}$$

are locally finite in X.

Proof. There exists a nonempty open subspace $U \subset X$ such that s corresponds to a section of $\Gamma(U, \mathcal{L})$ which generates $\mathcal{L}$ over U. Hence the prime divisors which can occur in the sets of the lemma all correspond to irreducible components of $|X| \setminus |U|$. Hence Lemma 0EE5. gives the desired result. □

Date: stacks-project:spaces-over-fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.